

RETAIL CUSTOMER RETENTION ANALYTICS – TESCO

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Course Name: Data Science using AI

INTRODUCTION

Retail businesses need to understand customer behaviour in order to improve their products and services. By analyzing customer data, businesses can identify purchasing patterns, understand customer preferences, and evaluate how different factors influence customer retention. This project focuses on analyzing important aspects such as customer churn, purchasing behaviour, loyalty programs, and customer lifetime value.

Power BI is used in this project as the main tool for data analysis and visualization. By creating interactive dashboards and visual reports, the data can be explored more effectively. These insights help businesses understand customer trends and support better decision-making for improving customer retention and overall business performance.

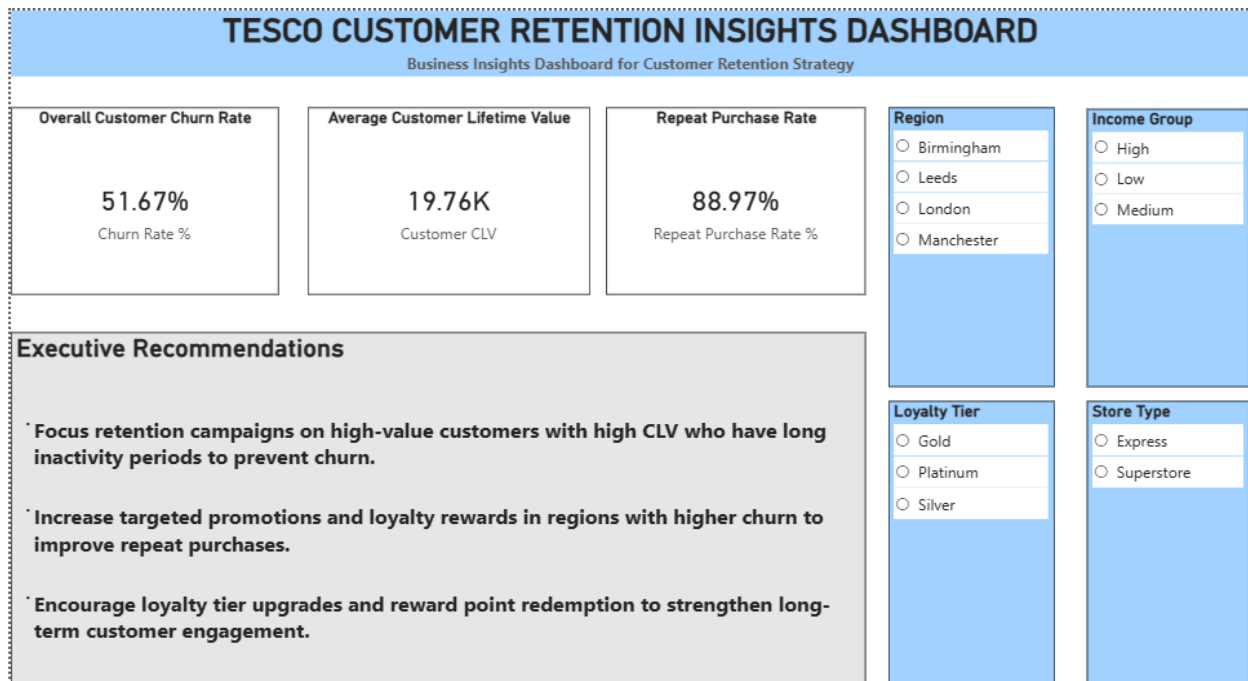


Figure 1 : Final Power Bi Customer Retention Dashboard

PROBLEM STATEMENT

- Customer retention is very important for retail businesses because keeping existing customers is often more cost-effective than acquiring new ones. When customers stop purchasing from a business, it leads to customer churn, which can negatively impact revenue and long-term growth.
- Churn increases marketing costs because companies must spend more resources to attract new customers. Therefore, businesses need to understand the reasons behind customer churn and identify patterns in customer behavior.
- Through data analysis, raw customer data can be transformed into meaningful insights. These insights help businesses understand customer behavior, identify areas for improvement, and develop strategies to increase customer retention and long-term growth.

TASK 1: DATA MODELING AND CLEANING

- Before starting the analysis, the dataset was prepared using the Power Query Editor in Power BI. Cleaning and transforming the data was important to make sure everything was accurate and ready for use.

The screenshot displays the Power Query Editor window for a query named 'Tesco_Customer_Retention'. The interface includes a ribbon with tabs: Home, Transform, Add Column, View, Tools, and Help. The 'Home' tab is active, showing options like 'Close & Apply', 'New Source', 'Recent Sources', 'Enter Data', 'Data source settings', 'Manage Parameters', 'Export query results', 'Refresh Preview', 'Advanced Editor', 'Choose Columns', 'Remove Columns', 'Keep Rows', 'Remove Rows', 'Sort', 'Split Column', 'Group By', 'Use First Row as Headers', 'Replace Values', 'Merge Queries', 'Append Queries', 'Combine Files', and 'Combine'. The main area shows a table with columns: Transaction_ID, Customer_ID, Transaction_Date, Store_ID, Product_Category, Amount, and Quantity. Each column has a data type icon and a status bar indicating the percentage of valid, error, and empty data. The 'Applied Steps' pane on the right shows the following steps: Source, Promoted Headers, Changed Type, Removed Duplicates, and Changed Type1. The 'Properties' pane on the right shows the query name 'Customer_Transactions'.

Transaction_ID	Customer_ID	Transaction_Date	Store_ID	Product_Category	Amount	Quantity
1	T20000	C1011	11/11/2024	S106	Bakery	249.6
2	T20001	C1079	5/16/2025	S105	Beverages	431.22
3	T20002	C1215	8/9/2024	S101	Beverages	233.28
4	T20003	C1263	8/16/2024	S108	Beverages	470.27
5	T20004	C1148	1/2/2025	S103	Electronics	254.48
6	T20005	C1080	7/5/2024	S106	Grocery	474.85
7	T20006	C1214	1/25/2025	S103	Bakery	417.25
8	T20007	C1197	9/27/2024	S107	Electronics	184.4
9	T20008	C1199	6/7/2025	S107	Clothing	432.5
10	T20009	C1024	6/9/2025	S109	Bakery	59.48
11	T20010	C1088	3/17/2025	S105	Bakery	462.9
12	T20011	C1267	6/18/2024	S108	Bakery	317.66
13	T20012	C1142	8/15/2024	S110	Beverages	31.92
14	T20013	C1058	5/31/2025	S110	Bakery	323.91
15	T20014	C1025	4/18/2025	S108	Grocery	487.25
16	T20015	C1046	8/23/2024	S101	Bakery	39.55
17	T20016	C1287	12/21/2024	S105	Beverages	279.57
18	T20017	C1265	7/29/2024	S103	Bakery	375.16
19	T20018	C1015	6/2/2025	S101	Clothing	406.08
20	T20019	C1198	10/20/2024	S104	Clothing	209.84
21	T20020	C1281	4/24/2025	S108	Beverages	106.06
22	T20021	C1085	3/1/2025	S103	Grocery	25.96
23	T20022	C1249	11/4/2024	S110	Grocery	291.96

Figure 2 : Data Cleaning using Power Query Editor (Applied Steps)

- First, duplicate records were checked and removed using key columns like Customer_ID and Store_ID. This step helps avoid wrong results that can happen due to repeated data.
- Next, the data types of different columns were corrected. Date columns such as Transaction_Date and Membership_Since were converted into proper date format. Numeric columns like Amount, Quantity, and Points were also checked to make sure they were stored as numbers, so calculations would work correctly.

Figure 3: Data type Formatting in Power Query

- Missing or null values were also handled carefully. Wherever data was incomplete, filtering or replacement methods were used to keep the dataset clean and reliable.

CALCULATED COLUMNS

Some extra columns were created to make the analysis easier and more meaningful.

MEMBERSHIP DURATION

A new column called Membership_Duration was created to find out how long each customer has been part of the loyalty program.

FORMULA USED:

Membership_Duration = Today() – Membership_Since

This helps understand how long customers have stayed with the business.

TRANSACTION YEAR AND TRANSACTION MONTH

Two more columns were created from the Transaction_Date column:

- Transaction_Year – shows the year of each transaction
- Transaction_Month – shows the month of each transaction These columns help in analyzing trends over time.

Table: Customer_Demographics (300 rows) Column: Membership_Duration (5 distinct values)

Customer_ID	Gender	Age	Membership_Since	Marital_Status	Region	Income_Group	Membership_Duration	Age_Group	CLV_Segment
C1000	Male	50	Sunday, November 1, 2020	Single	London	High		6 Senior	Low Value
C1002	Male	36	Wednesday, August 18, 2021	Single	Birmingham	Medium		5 Adult	Low Value
C1005	Female	61	Thursday, November 19, 2020	Single	Liverpool	High		6 Senior	Low Value
C1010	Male	21	Wednesday, May 3, 2023	Single	Manchester	Medium		3 Young	Low Value
C1017	Female	51	Tuesday, July 11, 2023	Single	London	High		3 Senior	Medium Value
C1018	Female	23	Thursday, March 21, 2024	Single	Liverpool	Low		2 Young	Low Value
C1019	Male	39	Wednesday, June 7, 2023	Single	Leeds	Low		3 Adult	Low Value
C1022	Female	33	Monday, December 19, 2022	Single	London	Low		4 Adult	Medium Value
C1026	Female	33	Friday, November 24, 2023	Single	Birmingham	Medium		3 Adult	High Value
C1027	Female	46	Wednesday, September 8, 2021	Single	Birmingham	High		5 Adult	Low Value
C1028	Female	20	Monday, August 30, 2021	Single	Liverpool	Low		5 Young	Medium Value
C1031	Male	36	Sunday, November 26, 2023	Single	Manchester	Medium		3 Adult	High Value
C1033	Female	20	Tuesday, August 2, 2022	Single	Birmingham	High		4 Young	High Value
C1034	Female	36	Saturday, August 26, 2023	Single	Birmingham	Low		3 Adult	Low Value
C1037	Male	24	Wednesday, April 5, 2023	Single	London	Low		3 Young	Low Value
C1038	Male	69	Saturday, February 10, 2024	Single	London	Medium		2 Senior	High Value
C1042	Female	56	Monday, February 26, 2024	Single	London	Low		2 Senior	High Value
C1047	Male	45	Sunday, October 8, 2023	Single	London	High		3 Adult	Low Value
C1049	Female	67	Thursday, September 14, 2023	Single	London	Low		3 Senior	Medium Value
C1056	Female	33	Friday, September 10, 2021	Single	Leeds	Low		5 Adult	Low Value
C1059	Male	69	Thursday, June 30, 2022	Single	Birmingham	High		4 Senior	Medium Value
C1061	Male	19	Tuesday, June 29, 2021	Single	London	Low		5 Young	Low Value
C1063	Male	65	Monday, December 26, 2022	Single	Liverpool	High		4 Senior	Low Value
C1064	Male	29	Thursday, September 7, 2023	Single	Liverpool	Medium		3 Young	High Value
C1065	Female	22	Saturday, October 1, 2022	Single	London	Medium		4 Young	Medium Value
C1066	Female	54	Tuesday, August 4, 2020	Single	Birmingham	High		6 Senior	Medium Value
C1067	Male	49	Friday, May 10, 2024	Single	Birmingham	Medium		2 Adult	High Value

Table: Customer_Transactions (290 rows) Columns: Transaction_Year (2 distinct values)

Transaction_ID	Customer_ID	Transaction_Date	Store_ID	Product_Category	Amount	Quantity	Promotion_Applied	Transaction_Year	Transaction_Month	Column
T20000	C1011	Monday, November 11, 2024	S106	Bakery	249.6	7	Yes	2024	11	
T20006	C1214	Saturday, January 25, 2025	S103	Bakery	417.25	2	Yes	2025	1	
T20009	C1024	Monday, June 9, 2025	S109	Bakery	59.48	3	Yes	2025	6	
T20010	C1088	Monday, March 17, 2025	S105	Bakery	462.9	4	No	2025	3	
T20011	C1267	Friday, June 28, 2024	S108	Bakery	317.66	1	No	2024	6	
T20013	C1058	Saturday, May 31, 2025	S110	Bakery	323.91	10	Yes	2025	5	
T20015	C1046	Friday, August 23, 2024	S101	Bakery	39.55	2	Yes	2024	8	
T20017	C1265	Monday, July 29, 2024	S103	Bakery	375.16	8	No	2024	7	
T20025	C1006	Sunday, April 13, 2025	S110	Bakery	35	7	Yes	2025	4	
T20032	C1272	Thursday, October 24, 2024	S104	Bakery	455.56	9	No	2024	10	
T20038	C1097	Monday, July 1, 2024	S101	Bakery	323.08	7	No	2024	7	
T20039	C1130	Tuesday, December 3, 2024	S109	Bakery	43.89	2	Yes	2024	12	
T20043	C1038	Sunday, April 27, 2025	S104	Bakery	352.81	9	No	2025	4	
T20045	C1003	Thursday, March 20, 2025	S107	Bakery	119.24	7	No	2025	3	
T20051	C1169	Saturday, January 11, 2025	S104	Bakery	253.28	10	Yes	2025	1	
T20053	C1250	Saturday, January 4, 2025	S104	Bakery	343.31	2	No	2025	1	
T20065	C1271	Tuesday, June 24, 2025	S105	Bakery	68.33	5	Yes	2025	6	
T20067	C1238	Tuesday, February 18, 2025	S106	Bakery	220.52	5	Yes	2025	2	
T20070	C1176	Wednesday, December 11, 2024	S104	Bakery	446.49	2	No	2024	12	
T20072	C1141	Wednesday, July 3, 2024	S107	Bakery	278.18	10	Yes	2024	7	
T20080	C1186	Saturday, November 16, 2024	S104	Bakery	222.04	8	No	2024	11	
T20087	C1225	Friday, January 10, 2025	S103	Bakery	107.6	3	Yes	2025	1	
T20090	C1051	Monday, November 11, 2024	S106	Bakery	15.16	4	Yes	2024	11	
T20096	C1242	Thursday, January 23, 2025	S103	Bakery	89.21	1	Yes	2025	1	
T20099	C1217	Wednesday, July 3, 2024	S101	Bakery	42.86	2	No	2024	7	
T20104	C1260	Tuesday, November 12, 2024	S101	Bakery	354.8	6	Yes	2024	11	

Figure 4 and 5 : Created Calculated Columns

DATA MODEL RELATIONSHIPS

After cleaning the data, relationships between different tables were created in the Data Model View.

The relationships are:

- Customer_Demographics → Transactions (One-to-Many)
- Customer_Demographics → Loyalty_Program (One-to-Many)
- Customer_Demographics → Churn_Labelled_Customers (One-to-Many)
- Transactions → Store_Locations (Many-to-One)

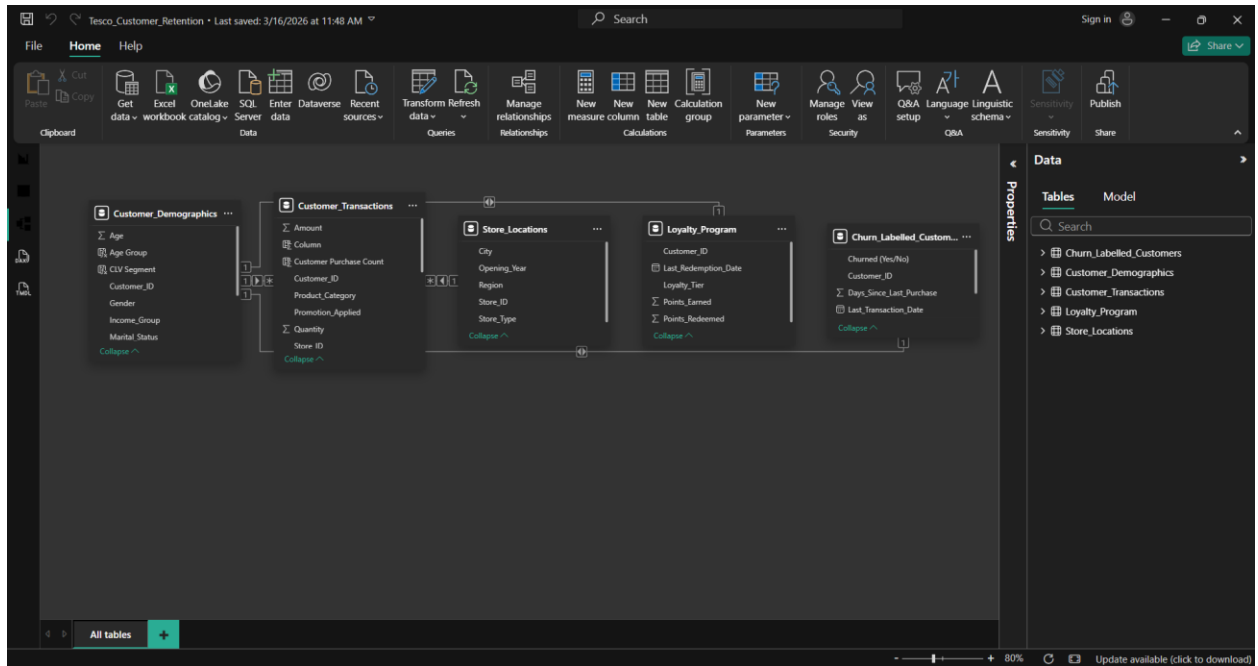


Figure 6 : Data Model Relationships

TASK 2: CHURN AND RETENTION METRICS

The goal of this task is to study customer churn and understand how it changes across different groups of customers. This helps the business find ways to improve customer retention.

Churn Rate Calculation

A churn rate metric was created to measure how many customers have stopped purchasing.

Formula used:

$$\text{Churn Rate} = (\text{Churned Customers} \div \text{Total Customers}) \times 100$$

The overall churn rate is 51.67%, which means more than half of the customers are no longer active.

Churn Rate Analysis

To understand churn better, it was analyzed based on different customer segments:

- Region
- Store Type
- Income Group

The analysis shows that churn rate is almost the same across different regions and store types. However, there is a small difference based on income group, where high-income customers have slightly higher churn compared to others.

Funnel Analysis

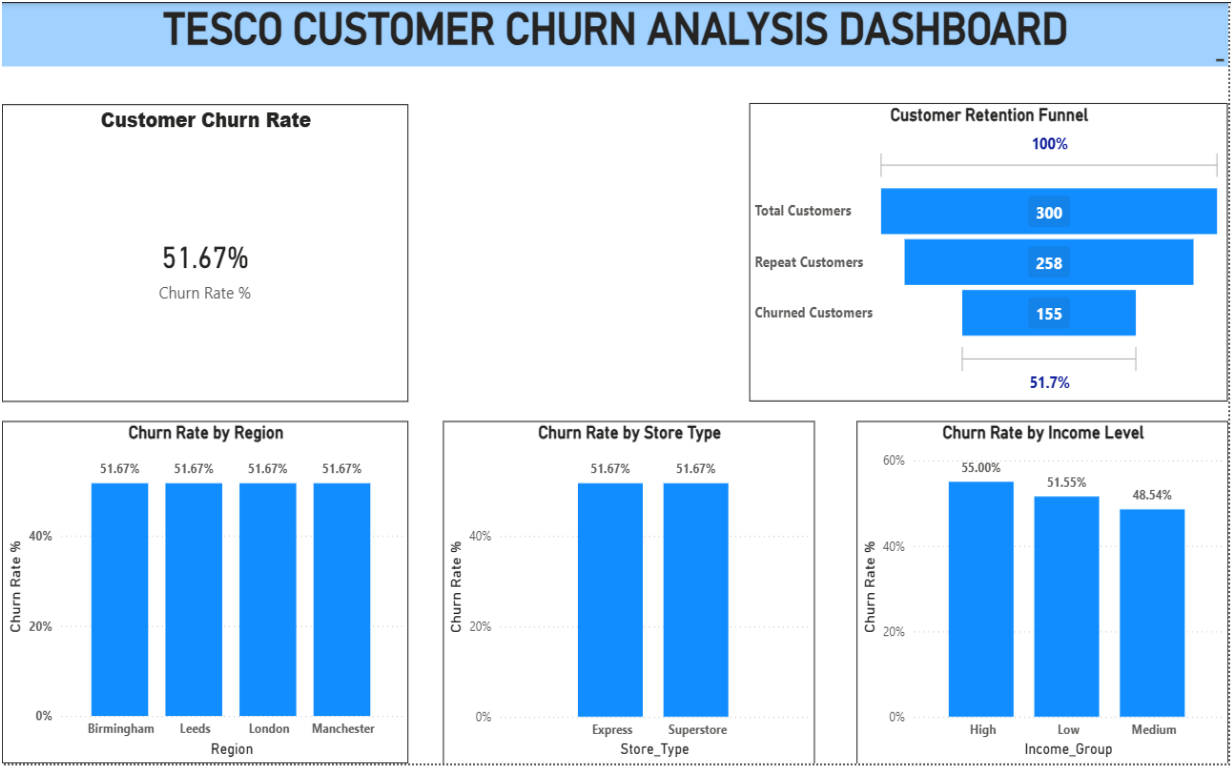
A funnel chart was used to show the customer journey:

- Total Customers: 300
- Repeat Customers: 258
- Churned Customers: 155

This indicates that even though many customers return and make repeat purchases, a large number of them eventually stop buying, which leads to a high churn rate.

Key Insight

The results show that churn is spread across all segments and is not limited to any specific region or store type. This means the business should focus on overall customer retention strategies instead of targeting only one group.



TASK 3: REPEAT PURCHASE ANALYSIS

The aim of this task is to study how customers make repeat purchases and understand their loyalty. This helps in knowing how often customers come back and buy again.

Customer Segmentation Based on Purchase Frequency

Customers were divided into different groups based on how many purchases they made:

- Low-Tier Customers: 2 to 4 purchases
- Mid-Tier Customers: 5 to 10 purchases
- High-Tier Customers: More than 10 purchases

Purchase Frequency Analysis

The average purchase frequency was studied across different categories:

- Region
- Age Group
- Loyalty Tier

The results show that the average purchase frequency is almost the same across all these segments. This means customers behave in a similar way regardless of their region, age, or loyalty level.

Product Category Analysis

The analysis also looked at which product categories are bought the most.

From the dashboard, it is observed that:

- Bakery products have the highest number of purchases
- Followed by beverages and electronics

This shows that these categories are more popular and contribute more to overall sales.

Key Insight

The data shows that most customers have a similar purchase pattern, with very little difference in how often they buy. This suggests that customer loyalty is not strongly varied in this dataset.

Because of dataset limitations, the variation in repeat purchases is low. However, the segmentation still helps demonstrate how customers can be grouped based on their buying behavior.

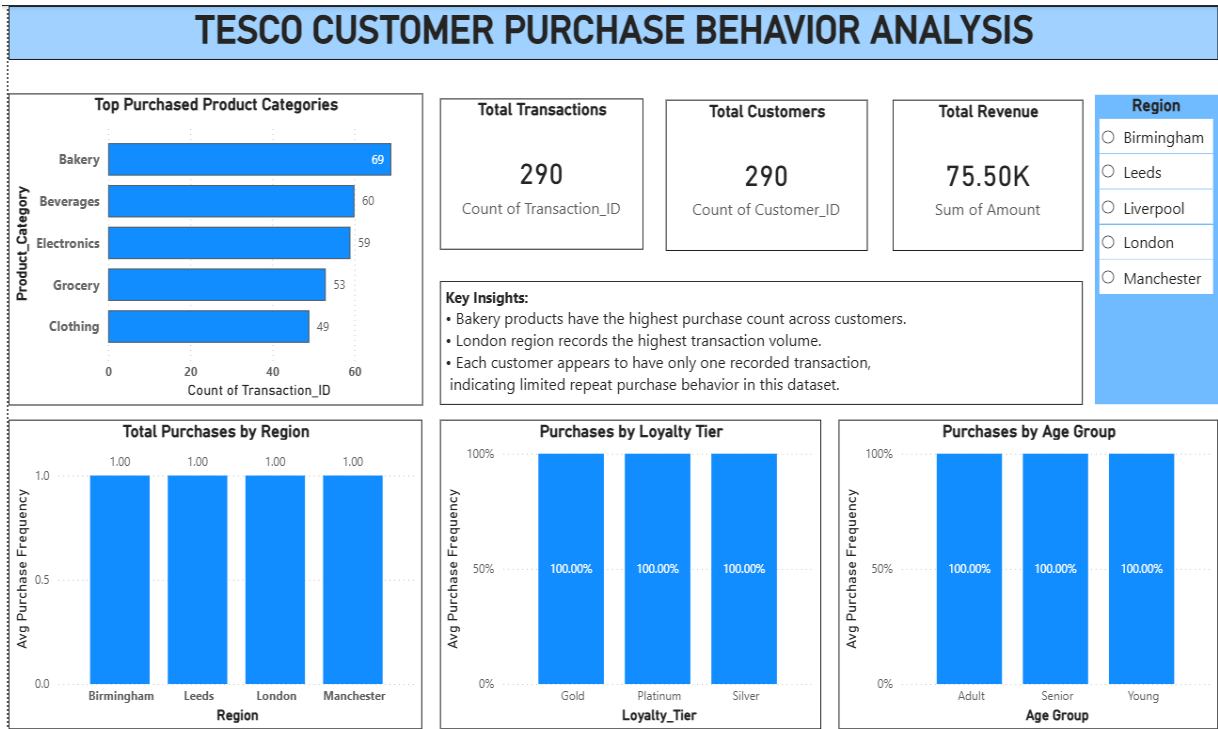


Figure 8 : Customer Purchase Behaviour Analysis

TASK 4: PROMOTION & LOYALTY IMPACT

The purpose of this task is to understand how promotions and loyalty programs affect customer behavior and retention. This helps in checking whether offers and rewards are actually encouraging customers to buy more and stay connected with the business.

Promotion Usage Analysis

A measure was created to find out how often promotions are used in transactions.

From the dashboard, the promotion usage rate is 50%, which means promotions are applied in half of the total transactions.

Purchase Behavior with Promotions

The average purchase amount was compared for transactions with and without promotions:

- With promotion: 253.23
- Without promotion: 267.46

This shows that purchases without promotions have a slightly higher average value. It suggests that while promotions may not increase the amount spent in a single purchase, they can still encourage customers to buy more frequently.

Churn Rate by Loyalty Tier

Customer churn was analyzed based on different loyalty levels:

- Platinum customers have the highest churn rate
- Silver customers show moderate churn
- Gold customers have the lowest churn rate

This indicates that even high-value customers (like Platinum) may need better engagement to keep them loyal.

Points Earned vs Redeemed

A comparison was made between points earned and points redeemed across loyalty tiers.

The analysis shows that:

- Customers earn more points than they redeem
- Redemption is low across all loyalty levels

This means many customers are not fully using the benefits of the loyalty program.

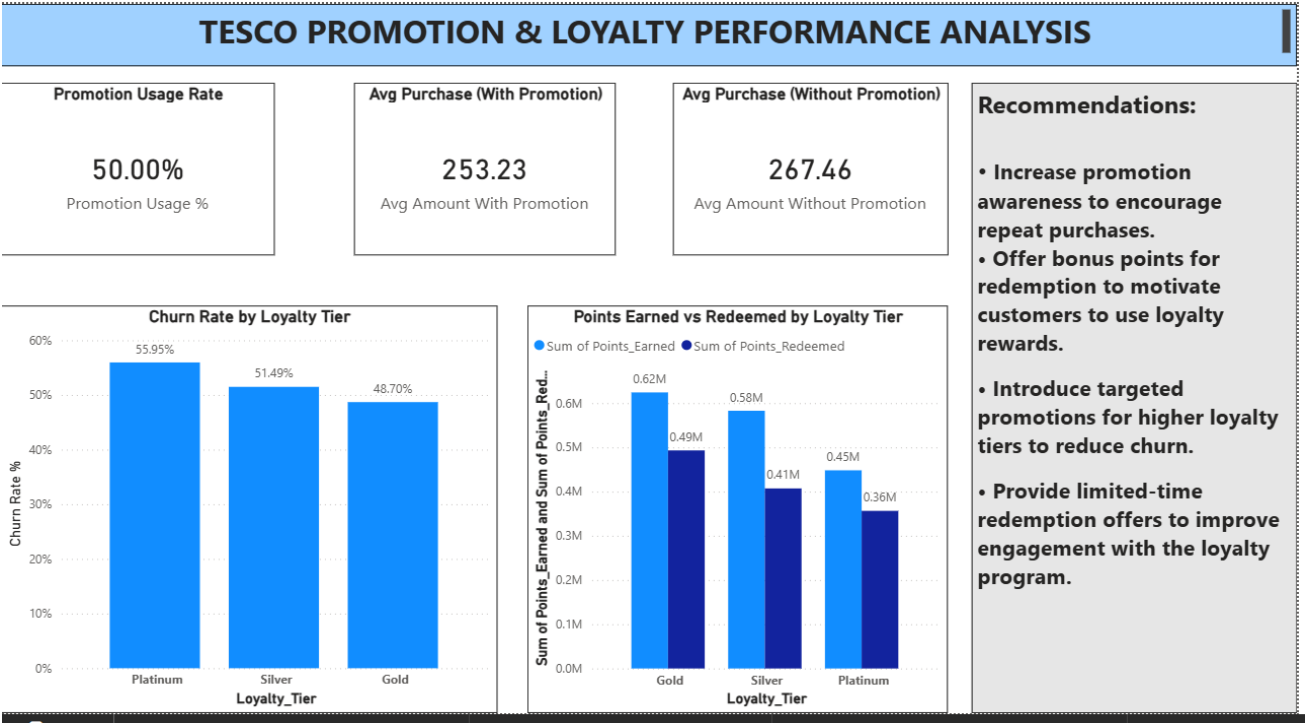
Key Insight

The results show that promotions are commonly used, but they do not greatly increase the average purchase value. Also, the gap between points earned and redeemed suggests that customers are not highly engaged with the loyalty program.

Recommendations

Based on the analysis, the following steps can help improve customer retention:

- Increase awareness of promotions to encourage repeat purchases
- Provide better rewards or bonus points to increase redemption
- Offer targeted promotions for higher loyalty tiers to reduce churn
- Use limited-time offers to improve engagement



TASK 5: STORE PERFORMANCE VS RETENTION

The aim of this task is to understand how different store features affect customer retention. This helps identify whether certain stores perform better and where improvements are needed.

Data Integration

To do this analysis, the Store_ID from the transactions data was connected with the Store_Locations table. This made it possible to include details like store type, region, and opening year in the analysis.

Average Transaction Amount by Store Type

The average amount spent per transaction was compared across different store types.

From the dashboard:

- Express stores have a slightly higher average transaction value
- Superstores have slightly lower values, but the difference is very small

This shows that both store types perform almost the same in terms of customer spending.

Churn Rate by Store Region

Churn rate was analyzed across different regions:

- Birmingham
- Leeds
- London
- Manchester

The results show that churn rate is almost the same in all regions. This means customer churn is consistent and not limited to any specific location.

Store Opening Year vs Retention

A scatter plot was used to study the relationship between store opening year and customer retention.

The analysis shows no major difference between older and newer stores. This means the age of the store does not have a strong impact on customer churn.

Key Insight

The results suggest that store-related factors like region, type, and opening year do not significantly affect customer churn. Customer behavior appears to be similar across all stores.

Recommendations

Based on the findings, the following actions can be taken:

- Focus on overall retention strategies instead of targeting specific regions
- Promote high-value products in both store types to increase revenue
- Strengthen loyalty programs, especially in newer stores
- Use regional promotions to keep customers engaged

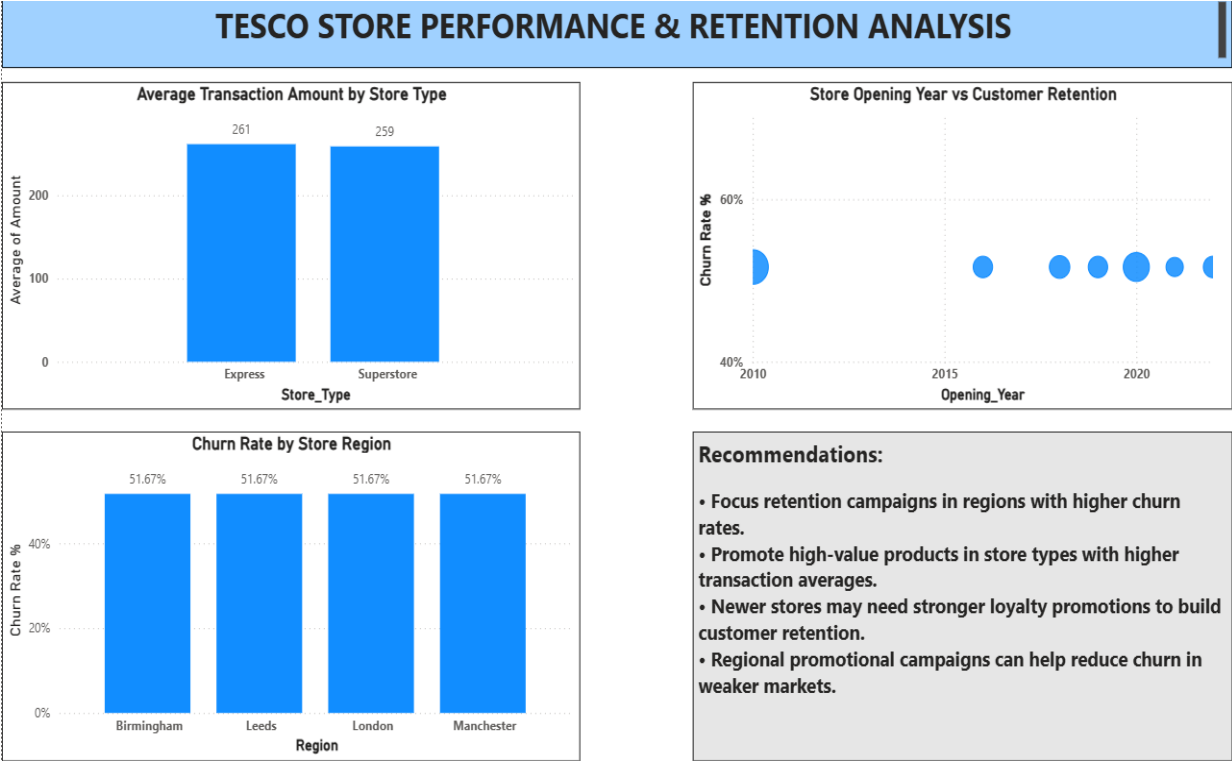


Figure 10 : Store Performance And Retention Analysis

TASK 6: CUSTOMER VALUE (CLV) ANALYSIS

The goal of this task is to identify high-value customers and understand how much they contribute to the business over time. Customer Lifetime Value (CLV) helps measure how valuable a customer is based on their spending and relationship with the company.

CLV Calculation

Customer Lifetime Value (CLV) was calculated using the formula:

$$\text{CLV} = \text{Total Amount Spent} \div \text{Membership Duration (in years)}$$

This helps in finding customers who bring more value to the business over a longer period.

Customer Segmentation

Customers were divided into three groups based on their CLV:

- Low Value Customers (Bottom 25%)
- Medium Value Customers (Middle 50%)
- High Value Customers (Top 25%)

This makes it easier to focus on the most valuable customers and plan better marketing strategies.

CLV vs Days Since Last Purchase

A scatter plot was used to study the relationship between CLV and how recently customers made a purchase.

From the analysis:

- High CLV customers are spread across different time gaps
- Some high-value customers have not made purchases recently

This shows that even valuable customers can become inactive and may need attention.

CLV by Loyalty Tier and Region

CLV was compared across different loyalty levels and regions.

The results show that:

- Some regions have higher CLV than others
- Higher loyalty tiers tend to have better CLV, showing stronger customer engagement

This helps identify which regions and customer groups are performing well.

Key Insight

The analysis shows that high-value customers are not always the most active. Some of them have not been purchased recently, which means there is a risk of losing them.

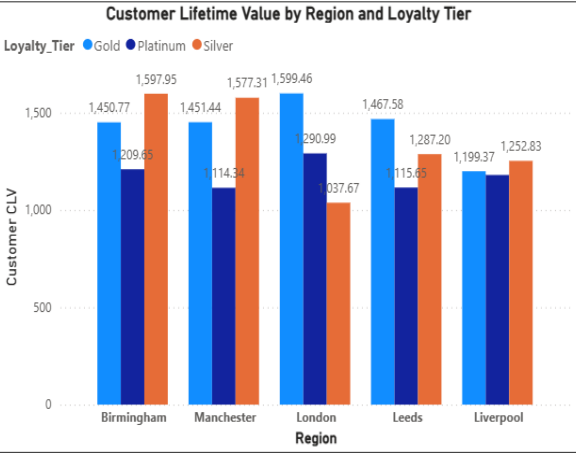
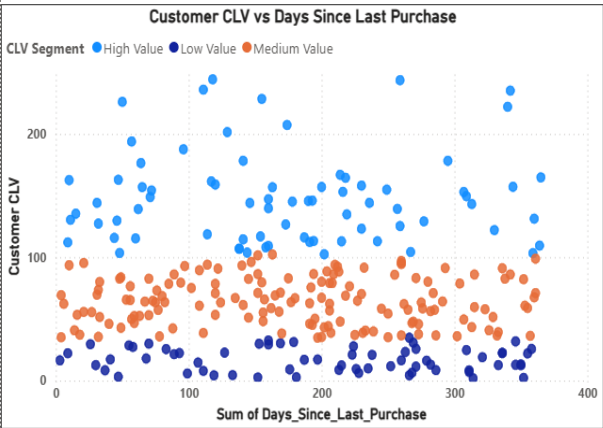
Recommendations

Based on the findings, the following actions are suggested:

- Focus on retaining high CLV customers through targeted campaigns
- Re-engage inactive high-value customers with personalized offers
- Pay more attention to regions with higher CLV to increase revenue
- Improve loyalty programs for medium-value customers to move them into the high-value group

TESCO CUSTOMER VALUE & RETENTION ANALYSIS

Customer CLV	Customer_ID	CLV Segment
12.52	C1000	Low Value
15.67	C1001	Low Value
39.00	C1002	Low Value
28.35	C1003	Medium Value
98.77	C1004	Medium Value
8.72	C1006	Low Value
52.45	C1007	Medium Value
36.92	C1008	Medium Value
19,764.37		



Key Insights:

- High CLV customers should be prioritized for loyalty campaigns.
- Regions with higher CLV customers represent high-value markets.
- Customers with high CLV but long inactivity periods should receive targeted retention offers.
- Loyalty tiers with higher CLV indicate effective engagement strategies.

figure 11 : Customer Value And Retention Analysis

TASK 7: FINAL DASHBOARD & EXECUTIVE SUMMARY

Objective

The aim of this task is to bring together all the previous analysis into one clear and easy-to-use dashboard. This dashboard helps business users quickly understand important metrics, customer behavior, and make better decisions to improve customer retention.

Multi-Page Report Structure

A multi-page Power BI report was created to organize the insights in a simple and structured way:

Page 1: Overview KPIs

This page shows the most important metrics:

- Customer Churn Rate
- Average Customer Lifetime Value (CLV)
- Repeat Purchase Rate

These KPIs give a quick summary of overall business performance.

Page 2: Loyalty & Promotion Impact

This page focuses on how promotions and loyalty programs affect customers:

- Promotion usage rate
- Average purchase with and without promotions
- Churn rate across loyalty tiers
- Points earned vs redeemed

This helps understand whether loyalty programs are working effectively.

Page 3: Store and Region Insights

This page looks at store and regional performance:

- Average transaction amount by store type
- Churn rate by region
- Impact of store opening year on retention

This helps identify which locations are performing well and where improvements are needed.

Page 4: Customer Segmentation

This page focuses on customer behavior and value:

- Segmentation into Low, Medium, and High-value customers
- CLV analysis
- Identification of churned, repeat, and high-value customers

This helps businesses focus on the most important customers.

Use of Slicers

Interactive slicers were added to make the dashboard more flexible and user-friendly:

- Region
- Income Group
- Loyalty Tier
- Store Type

These allow users to filter the data and explore insights based on different conditions.

Executive Summary

The dashboard highlights some key findings:

- The churn rate is quite high, showing the need for better retention strategies
- Repeat purchases are strong, which means there is good potential for long-term relationships
- Customer Lifetime Value varies across regions and loyalty levels

Key Recommendations

- Focus more on high-value customers and keep them engaged
- Improve loyalty programs by encouraging customers to use their points
- Target areas with higher churn using better promotions and campaigns

Business Focus Areas

TESCO should mainly focus on:

- Retaining high-value customers
- Increasing engagement through loyalty programs
- Improving performance in regions with higher churn

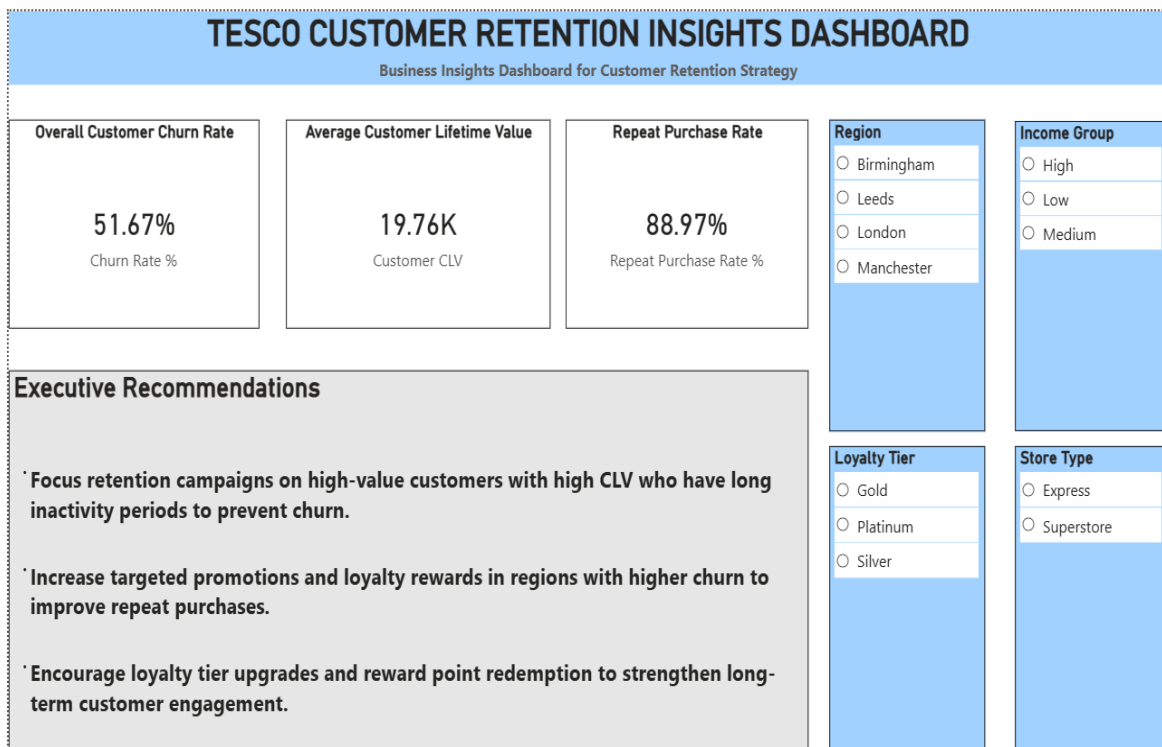


Figure 12 : Customer Retention Insights Dashboard

TASK 8: VIDEO EXPLANATION

A video presentation has been prepared to clearly explain the overall project. It covers the introduction, problem statement, key analysis, important insights, and final business recommendations.

This video helps in understanding the project in a simple and visual way.

Video Link: <https://drive.google.com/file/d/1YEEwpzPCNgTh3JfBLi-whQmh4TtR0miW/view?usp=sharing>